

MATH 211 Analysis III — Problem Session 3

Problem 1

Decide which of the following sets are compact:

$$S_0 = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} \cup \{0\}, \quad S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Solution. We use the open cover definition of compactness. A set X in $\mathbb{R}^n$ is compact if every open cover of X has a finite subcover. Let $\bigcup_{\alpha \in I} A_\alpha$ where I is an index set be an open cover of S_0 . Since 0 is inside S_0 there exists some A_{α_0} that covers the neighborhood around 0, say A_{α_0} covers $\text{Br}(0, \epsilon)$ for some $\epsilon > 0$. Observe that A_{α_0} covers all the points $\frac{1}{n}$ in S_0 satisfying $n > \frac{1}{\epsilon}$. Let $m = \lfloor \frac{1}{\epsilon} \rfloor$. Thus, $A_{\alpha_0} \cup A_{\alpha_1} \cup A_{\alpha_2} \cup \dots \cup A_{\alpha_m}$ where $A_{\alpha_1}, \dots, A_{\alpha_m}$ are open sets of the cover that cover $1, \frac{1}{2}, \dots, \frac{1}{m}$ covers S_0 . Thus, S_0 is compact. We note that some of the A_{α_i} 's in the finite cover could be the same sets, you can form a smaller cover by taking the repeated sets once.

Exercise. The argument in this solution does not apply to S because S does not contain 0. Show that S is not compact. You may show that if you take an open cover of S and take any finite subfamily of it then there always will be points in S that will not be covered.

We recall the **Heine–Borel Theorem**:

Theorem (Heine–Borel). A set X in $\mathbb{R}^n$ is compact if and only if X is bounded and closed.

Problem 2

Explain why the following sets are not compact:

$$A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}, \quad B = \{(x, y) \in \mathbb{R}^2 : y \geq 0\}.$$

Solution. A^c is not open because the points (x, y) where $x^2 + y^2 = 1$ are in A^c and any open ball around such (x, y) points is not contained in A^c , since it contains points of A . Thus, A is not closed. By Heine–Borel theorem A is not compact.

B is closed (Why?) but not bounded. Indeed, suppose B is bounded by some real number M , then since point $(M + 1, 0) \in B$, $\|(M + 1, 0)\| = M + 1 > M$ is a contradiction. By Heine–Borel theorem B is not compact.

Problem 3

Show that the set

$$K = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

is compact.

Solution. From previous weeks, we know that K is closed and K is bounded by 1. By Heine–Borel theorem K is compact.

Problem 4

Find the domain and range of $f(x, y) = \sqrt{1 - x^2 - y^2}$.

Solution. We are finding the largest domain and largest range that f could have. Since root function is defined for only non-negative values, $1 - x^2 - y^2 \geq 0$ which implies $x^2 + y^2 \leq 1$. So the largest domain of f is the closed unit disk around origin. More precisely,

$$\text{Domain}(f) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

Now in order to find the range, say $t = x^2 + y^2$. Note that $0 \leq t \leq 1$ and we solve the equation $f(x, y) = c$ for $c \geq 0$ (a square root is never negative).

$$\sqrt{1-t} = c \iff 1-t = c^2 \iff t = 1-c^2.$$

Since $0 \leq t \leq 1$, we need $0 \leq 1-c^2 \leq 1$, which holds if and only if $c \in [0, 1]$.

Thus, $\text{Range}(f) = [0, 1]$.

Problem 5

Find the domain and range of $f(x, y) = \ln(x - y)$.

Solution. Since $\ln$ function is defined only for positive values, $x - y > 0$. Thus, domain of f is

$$\text{Dom}(f) = \{(x, y) \in \mathbb{R}^2 : y < x\}.$$

To find range we solve the equation $\ln(x - y) = c$.

$$\ln(x - y) = c \iff x - y = \exp c \iff x = y + e^c.$$

Thus taking $x = y + e^c$ gives $f(x, y) = c$. Note that since e^c is always positive $x = y + e^c > y$. Hence for every $c \in \mathbb{R}$ the point $(y + e^c, y)$ is in the domain and f takes the value c there, so $\text{Range}(f) = \mathbb{R}$.

Problem 6

Find the domain and range of $f(x, y) = \frac{1}{x^2 + y^2 - 4}$.

Solution. The function is defined exactly when the denominator is nonzero, so the domain is $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \neq 4\}$, i.e. $\mathbb{R}^2$ minus the circle of radius 2.

For the range, we solve $f(x, y) = c$. If $c = 0$ there is no solution, since $\frac{1}{x^2 + y^2 - 4} \neq 0$ in the domain. If $c \neq 0$, the equation is equivalent to

$$x^2 + y^2 = 4 + \frac{1}{c}.$$

This has a solution if and only if $4 + \frac{1}{c} \geq 0$, and any solution lies in the domain because $\frac{1}{c} \neq 0$. This implies the range of f is

$$\left(-\infty, -\frac{1}{4}\right] \cup (0, \infty).$$

To observe this, one needs to find intervals where above inequality is satisfied (find critical points, determine the positive intervals etc.).

Exercise. Find the domain and range of $f(x, y) = \arcsin(x + y)$ and sketch the domain.

Problem 7

Sketch the graph of $f(x, y) = 6 - 2x - 3y$.

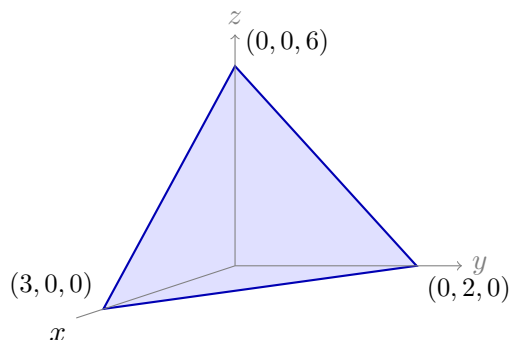
Solution. Let $z = 6 - 2x - 3y$. We first find the intercepts:

If $x = y = 0$ then the graph passes through $(0, 0, 6)$.

If $x = z = 0$ then the graph passes through $(0, 2, 0)$.

If $y = z = 0$ then the graph passes through $(3, 0, 0)$.

Connecting the points we get the graph:



This figure only shows the graph of the function in the first octant of xyz -space. This shape also extends to infinity in every direction.

Problem 8

Compute

$$(a) \lim_{(x,y) \rightarrow (1,2)} \frac{x^2y + 3xy}{x + y}, \quad (b) \lim_{(x,y) \rightarrow (1,1)} \frac{x^2 - y^2}{x - y}.$$

Solution. (a) The limits of the numerator and the denominator exist:

$$\lim_{(x,y) \rightarrow (1,2)} (x^2y + 3xy) = 1^2 \cdot 2 + 3 \cdot 1 \cdot 2 = 8, \quad \lim_{(x,y) \rightarrow (1,2)} (x + y) = 1 + 2 = 3 \neq 0.$$

Since the limit of the denominator is nonzero and f is continuous, the limit of the quotient is the quotient of the limits:

$$\lim_{(x,y) \rightarrow (1,2)} \frac{x^2y + 3xy}{x + y} = \frac{\lim_{(x,y) \rightarrow (1,2)} (x^2y + 3xy)}{\lim_{(x,y) \rightarrow (1,2)} (x + y)} = \frac{8}{3}.$$

Problem 9

Compute $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$.

Solution. We approach $(0, 0)$ along the x -axis.

$$\lim_{(x,0) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x^2 - 0^2}{x^2 + 0^2} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1.$$

Now we approach $(0, 0)$ along the y -axis,

$$\lim_{(0,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = \lim_{y \rightarrow 0} \frac{0^2 - y^2}{0^2 + y^2} = \lim_{y \rightarrow 0} \frac{-y^2}{y^2} = -1.$$

The limits along two paths differ, so the limit does not exist.

Problem 10

Compute $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$.

Solution. Since $x^2 \leq x^2 + y^2$,

$$0 \leq \left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{x^2}{x^2 + y^2} |y| \leq |y|, \quad \lim_{(x,y) \rightarrow (0,0)} |y| = 0,$$

so by the squeeze theorem

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0.$$